

# Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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## Abstract

Prepending a *primer* of factual sentences about a graph’s structure (node degree, clustering coefficient, random-walk return probabilities or component count) to its text encoding is a natural way to assist a language model on graph-reasoning questions. An apparent gain is uninformative if the answer can be read from the primer without consulting the graph, and confounded if the primer merely lengthens the prompt. We evaluate primers against four controls jointly: a graph-blind shortcut solver refit at the evaluated graph size, a length-matched content-free filler, a minimal-length content control, and a degree-preserving rewiring that varies triangle count at fixed prompt length. Across four Qwen3 arms (1.7B and 4B, with and without thinking mode) on 40-node Erdős–Rényi graphs, primers that state the answer help a model at floor and harm a model at ceiling (cross-fitted interaction with baseline accuracy  $-11.0$  points, 95% CI  $[-17.1, -5.7]$ ), whereas primers that do not state it show no such relation. The pattern replicates on eleven held-out arms (0.6–14B, two families) but is not graded at intermediate baselines. The only effect of a primer that does not state the answer, clustering on `node_degree`, is small when pooled over densities ( $+1.9$  points) and concentrated at intermediate densities ( $+5.9$ , replicated on independent graphs); two candidate mechanisms for it are not supported.

## 1 Introduction

Fatemi et al. (2023) show that a large language model’s accuracy on graph-reasoning questions varies substantially with the choice of text encoding when the graph and question are held fixed. We ask a complementary question about the same benchmark: does prepending a short *primer* of derived, node-level facts to the graph encoding improve the model’s answers?

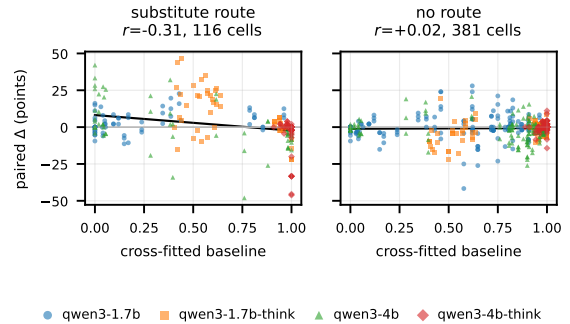


Figure 1: Cross-fitted paired effect against cross-fitted baseline accuracy (Section 5.3) for the 116 cells in which a primer offers a substitute route to the answer (left) and the 381 that do not (right), each shown once per fold direction. Baseline and effect come from disjoint halves of the paired graphs, so shared sampling noise cannot produce the slope.

Four confounds complicate the interpretation of such an effect. First, a primer may help because the model reads the answer from it rather than reasoning over the graph (answer leakage). Second, a primer changes prompt length, and context length and position alone affect retrieval and reasoning accuracy (Liu et al., 2024; Kamradt, 2023; Levy et al., 2024). Third, answer extraction can err in ways that correlate with response style, which appears as a spurious condition effect (Section 3.4). Fourth, a model near ceiling on a task leaves no room for any primer effect to appear.

We address three research questions. **RQ1:** when a primer improves accuracy, is the improvement attributable to the primer stating the answer? **RQ2:** does added text affect accuracy independently of its content? **RQ3:** does any primer that does not state the answer improve accuracy beyond a length-matched control, and through what mechanism? Our main contribution is a set of controls applied jointly: a graph-blind shortcut solver refit at the evaluated graph size (Section 3.2), a length-matched content-free filler, a minimal-length

content control (`components`), and a degree-preserving rewiring that changes triangle count at fixed prompt length (Section 3.3). We apply them to 40-node Erdős–Rényi graphs across four model arms and six tasks.

## 2 Related Work

We adopt the GraphQA task definitions of [Fatemi et al. \(2023\)](#) and manipulate graph size and edge density directly, so that difficulty is an experimental variable rather than an incidental one. [Wang et al. \(2023\)](#) report that LLMs are brittle to spurious correlations in graph problems and propose *Build-a-Graph* and *Algorithmic* prompting, the prompt-level interventions closest to a primer; [Perozzi et al. \(2024\)](#) instead learn a continuous graph encoding. [Guo et al. \(2023\)](#) find that models struggle to exploit graph description languages directly, and [Jin et al. \(2024\)](#) survey the wider literature. [Dziri et al. \(2023\)](#) argue that transformers solve compositional tasks by linearised subgraph matching rather than systematic reasoning, which raises the same question as a primer’s benefit: does it reflect reasoning or a shorter pattern to match? [Geirhos et al. \(2020\)](#) frame this as *shortcut learning*; our shortcut solver operationalises that concern for the primer channel. Irrelevant context degrades reasoning ([Shi et al., 2023](#)), as does input length well below the context limit ([Levy et al., 2024](#)); our `filler` condition measures this cost directly. The `rwse` condition adapts the random-walk structural encoding of [Dwivedi et al. \(2022\)](#), rendered as sentences, and the rewiring follows the double-edge-swap configuration model ([Fosdick et al., 2018](#)). For statistics we follow [Dror et al. \(2018\)](#) on paired testing, test differences between groups directly rather than contrasting their significance ([Gelman and Stern, 2006](#)), and borrow cross-fitting from [Chernozhukov et al. \(2018\)](#). We distinguish Qwen3’s native thinking mode, toggled through the chat template, from chain-of-thought *prompting* ([Wei et al., 2022](#)); this paper tests only the former.

## 3 Method

### 3.1 Primers

A primer is a preamble of factual sentences about a graph’s structure, prepended to the standard text encoding and the task question (Figure 2; a complete example appears in Appendix A.1). A single renderer produces all seven conditions, so conditions share sentence framing and number formatting and

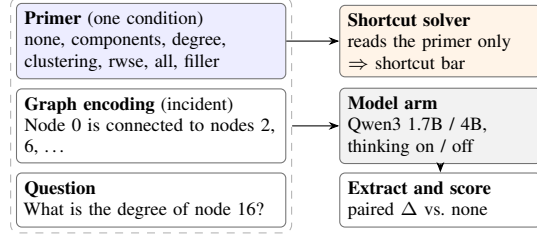


Figure 2: Experimental design. Each prompt joins a primer, the incident encoding and the question. The graph-blind shortcut solver reads only the primer and sets the bar a primer must exceed; `filler` controls length, `components` is a minimal-length content control, and rewiring (Section 3.3) changes the graph’s triangles while holding degrees and prompt length fixed.

differ only in the facts stated. The exception is **filler**, which states no fact and therefore uses its own frame (“Node  $N$  is simply present within the graph  $G$ .”); it is a length control, not a content control. **none** is the graph encoding alone. **components** gives one sentence stating the number of connected components. **degree** states each node’s degree, **clustering** each node’s clustering coefficient, and **rwse** each node’s random-walk return probabilities after two and three steps ([Dwivedi et al., 2022](#)). **all** combines degree, clustering and `rwse`. Mean primer length at  $n=40$  (Qwen3 tokenizer) is 8 tokens (`components`), 372 (`degree`), 470 (`filler`), 510 (`clustering`), 1,070 (`rwse`) and 1,612 (`all`), against a mean `none` prompt of 2,165 tokens. `filler` is thus a close, slightly conservative length control for `clustering`, but only bounds the length term for `rwse` and `all`.

### 3.2 Shortcut solvers

A primer’s measured benefit is evidence of graph reasoning only if it exceeds what a solver that never sees the graph can recover from the primer text. Our shortcut solver combines sixteen exact rules, one heuristic, eight fitted rules (fit and scored on disjoint graph sets) and an exact enumeration bound for small graphs. Its accuracy on a (task, condition) cell, the *shortcut bar*, is a content-only lower bound: it never overstates what the primer text alone permits.

A bar fit on the published 5–19-node graphs does not transfer to our  $n=40$  corpus, so we re-fit the solver on  $n=40$  Erdős–Rényi graphs at the seven densities of our sweep (disjoint 300-graph fit and test sets per density; Table 1). We classify a (task, condition) pair as offering a *substitute route* when its bar exceeds the `none` bar by more

| Task            | none | comp. | clust. | rwse | degree      | filler | all         |
|-----------------|------|-------|--------|------|-------------|--------|-------------|
| node_count      | 1.00 | 1.00  | 1.00   | 1.00 | 1.00        | 1.00   | 1.00        |
| edge_count      | .030 | .030  | .104   | .030 | <b>1.00</b> | .030   | <b>1.00</b> |
| node_degree     | .144 | .144  | .144   | .210 | <b>1.00</b> | .144   | <b>1.00</b> |
| connected_nodes | .000 | .000  | .000   | .001 | .001        | .000   | .008        |
| edge_existence  | .735 | .735  | .727   | .554 | .746        | .735   | .741        |
| cycle_check     | 1.00 | 1.00  | 1.00   | 1.00 | 1.00        | 1.00   | 1.00        |

Table 1: Shortcut bars at  $n=40$  (mean over seven densities): accuracy of a solver that reads only the rendered primer, never the graph. **Bold**: the primer content alone determines the answer. The unbolded 1.00 entries reflect a constant gold answer, not primer content.

than 0.05; every gain is either at most 0.011 or at least 0.066, so any threshold between these values yields the same split. Only `node_degree` (under `rwse`, `degree`, `all`) and `edge_count` (under `clustering`, `degree`, `all`) offer a route. `node_count` and `cycle_check` have constant gold answers at  $n=40$  (40, and “yes” at every density) and are excluded. The held-out arms, which answer the published split, use bars fit on that split (Appendix Table 4).

### 3.3 Rewiring: isolating structure from length

`filler` controls length only approximately, and comparisons across graph generators confound structure with class balance, so we use a within-instance design. A double-edge swap replaces edges  $(a, b), (c, d)$  with  $(a, d), (c, b)$  (Fosdick et al., 2018), which preserves the degree sequence,  $n, m$ , the `node_degree` gold answer and the prompt length while changing the triangle count that the `clustering` primer reports. We generate low, base and high variants of each graph ( $n=40$ , mean degree 12, 200 graphs per level; mean clustering 0.01, 0.31, 0.73) and evaluate `node_degree` under `none`, `clustering` and `filler` with `qwen3-1.7b`, `qwen3-1.7b-think` and `qwen35-2b`. If the `clustering` benefit derives from the stated values, it should vary with triangle count. If instead the model reads mean clustering as a density prior (in an Erdős–Rényi graph  $\bar{C} \approx p$ ), its degree estimates should shift toward  $\bar{C}(n-1)$ : 0.5, 12.0 and 28.7 across levels, against a true mean of 12.

### 3.4 Answer extraction and measurement validity

Extraction errors can masquerade as condition effects. On `cycle_check`, `qwen3-4b` often answers correctly and then recites that a cycle has “no repeated” nodes; a last-yes/no extractor mis-

read 225 such rows, all in one direction, producing a spurious collapse. All results use the corrected extractor, and a manual check of 50 rows per arm confirmed that `<think>` blocks never reach it. Truncated pairs are dropped; scoring them as incorrect changes the verdict on only three `qwen3-1.7b` `cycle_check` cells (Appendix A).

### 3.5 Significance testing

Every comparison is paired on the same graph and question and tested with the exact McNemar test (McNemar, 1947), Benjamini–Hochberg corrected (Benjamini and Hochberg, 1995) within each (arm, task) family at  $q=0.05$ ; a claim that a condition beats both controls also requires significance under a global correction. For the effect–baseline relation (Section 5.3), baseline and effect are estimated from disjoint random halves of each cell’s graphs, and the route  $\times$  baseline interaction is bootstrapped over (arm, task, density) blocks, which share one `none` sample. For null cells we simulate a minimum detectable effect (MDE) at power 0.8,  $\alpha=0.05$ .

## 4 Experimental Setup

**Tasks.** Six GraphQA tasks (Fatemi et al., 2023): node count, edge count, node degree, connected nodes, edge existence and cycle check.

**Models.** An *arm* is a model checkpoint with a fixed decoding configuration. The main arms are Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with native thinking mode off (`qwen3-1.7b`, `qwen3-4b`) and on (`-think`), chosen because they offer both modes of the same weights at two sizes and moderate cost.

**Sweep design.** All four arms answer one shared prompt set: Erdős–Rényi graphs (Erdős and Rényi, 1959) with  $n=40$  at densities  $p \in \{.10, .20, .35, .50\}$ , 100 graphs per (density, task) cell, seven conditions and six tasks, for 16,800 prompts per arm. Decoding is greedy with a budget of 2,048 new tokens for plain arms and 8,192 for thinking arms; Section 5.5 reports a budget-matched comparison. A high-density extension covers `node_degree` and `edge_existence` at  $p \in \{.65, .75, .85\}$  (100 graphs per cell, all four arms). For the one content effect that survives the controls (Section 5.4), a dedicated `node_degree` sweep with `qwen3-1.7b` uses 400 graphs per density at all seven densities, and a replication uses 400 newly seeded graphs per density at  $p \leq .50$ .

**Held-out arms.** Eleven further arms were not used to develop any claim reported here. Eight (Qwen3-8B, Qwen3-14B, Gemma-4-E4B-it and Gemma-4-12B-it (Google DeepMind, 2026), each with and without thinking) answer the published GraphQA split (30 graphs per cell, five tasks, six conditions); three (Qwen3-0.6B with and without thinking, and Qwen3.5-2B (Qwen Team, 2026), qwen3.5-2b) answer a 100-graph probe of none and degree on all six tasks. The published-split responses were collected before this project’s hypotheses were formed, so we describe these arms as held out rather than pre-registered.

**Metrics.** Exact match for every task (set-F1 on `connected_nodes` is 0.96–1.00 in every cell and does not discriminate); mean absolute error for the integer tasks is in the Appendix.

## 5 Results

### 5.1 Main sweep

Unless stated otherwise,  $\Delta$  is a paired difference against `none`, in percentage points, over instances in which neither condition was truncated. Table 2 (thinking arms in Table 3) pools over the four densities; per-cell bootstrap intervals are in the Appendix. On `edge_count` the thinking arms truncate on 62–81% of instances (Appendix Table 5), so those cells are reported but not interpreted; every other task truncates on fewer than 7%. Gains on the constant-answer tasks `node_count` and `cycle_check` are artifacts (Section 5.2). Several cells are significant in one arm but not the other; Section 5.3 shows why this is expected, since the two model sizes are rarely at matched baseline accuracy on the same cell.

### 5.2 Length is an active treatment

`filler` is not inert. It significantly reduces accuracy for qwen3-1.7b on `edge_existence` (−15.8) and `connected_nodes` (−6.2), and for qwen3-4b on both (−7.0, −12.5). Over the full seven-density range (qwen3-1.7b-think, `node_degree`) it costs 6.3 points pooled ( $p < 10^{-4}$ ): negligibly at  $p \leq .35$ , then 9.5, 9.3, 14.1 and 11.7 points at the four densest levels ( $r = +0.953$  with baseline across levels,  $p = 0.00088$ ). The cost of added length thus grows with task difficulty, in the same direction as the benefit of an answer-stating primer (Section 5.3).

The largest `edge_existence` effects are not changes in graph reading: the hit rate is 99–100%

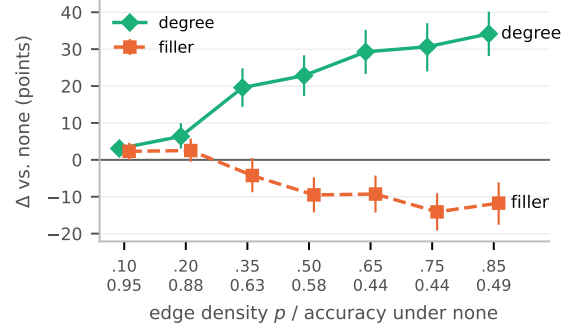


Figure 3: Density continuum, qwen3-1.7b-think, `node_degree`: paired effect against `none` (400 graphs per density, 95% bootstrap intervals). Lower tick row: accuracy under `none`.

under every condition, so accuracy moves only through the false-alarm rate, which `filler` raises and `clustering` lowers (Appendix Table 6). With hits at ceiling, a signal-detection analysis (Macmillan and Creelman, 2005; Hautus, 1995) cannot separate sensitivity from criterion (Appendix Table 7). On `node_count` (gold = 40), the dominant `none` error is an off-by-one (“39”, induced by the encoder’s “nodes 0, . . . , 39” preamble; 68.5% of qwen3-1.7b answers), which almost any interposed text reduces.

### 5.3 Answer-stating primers: route substitution

A *cell* is an (arm, task, density, condition) combination with at least ten usable pairs. On the main sweep plus the high-density extension, route cells gain at low baseline and lose at high baseline, while no-route cells do not (Figure 1; binned means in Appendix Table 8). Because baseline and effect share sampling noise, and a difference in significance is not a significant difference (Gelman and Stern, 2006), we test the cross-fitted route  $\times$  baseline interaction directly: −11.0 points per unit of baseline (95% block-bootstrap CI [−17.1, −5.7], 83 blocks; within groups  $r = -0.34$  for 116 route cells and  $r = +0.03$  for 381 no-route cells). The interaction persists on the logit scale, which removes the compression of a bounded outcome (−1.46 logits, CI [−2.11, −0.79]), but vanishes among cells with baseline between 0.10 and 0.90 (interaction CI [−29, +41]; Appendix Table 9). The evidence thus supports a contrast between floor and ceiling (a stated answer helps a model that cannot compute it and harms one that already can) rather than a graded dependence at intermediate baselines,

| Arm        | Task            | none | comp.        | clust.       | rwse         | degree       | filler       | all          |
|------------|-----------------|------|--------------|--------------|--------------|--------------|--------------|--------------|
| qwen3-1.7b | connected_nodes | 72.5 | +0.8         | +1.2         | +1.0         | +3.5         | <b>-6.2</b>  | +2.0         |
|            | cycle_check     | 99.7 | -0.6         | +0.3         | <b>-7.5</b>  | <b>-12.6</b> | +0.3         | -1.1         |
|            | edge_count      | 1.8  | -1.4         | -0.4         | -0.8         | +4.0         | -0.4         | +2.4         |
|            | edge_existence  | 70.5 | <b>-8.2</b>  | +3.8         | +5.3         | +4.0         | <b>-15.8</b> | <b>+11.5</b> |
|            | node_count      | 31.5 | <b>+21.2</b> | <b>+67.2</b> | +2.0         | <b>+6.8</b>  | <b>+39.8</b> | <b>+47.0</b> |
|            | node_degree     | 60.5 | +1.8         | +3.0         | +4.0         | <b>+7.5</b>  | -0.5         | <b>+10.2</b> |
| qwen3-4b   | connected_nodes | 86.0 | <b>+9.2</b>  | +1.0         | +0.0         | -3.0         | <b>-12.5</b> | -2.8         |
|            | cycle_check     | 99.7 | -1.3         | +0.3         | +0.3         | -1.5         | -0.3         | -0.5         |
|            | edge_count      | 2.0  | -1.3         | -1.3         | -1.5         | <b>+27.7</b> | +1.2         | <b>+3.8</b>  |
|            | edge_existence  | 90.5 | -2.8         | <b>+4.0</b>  | <b>-10.0</b> | <b>-4.3</b>  | <b>-7.0</b>  | <b>-9.8</b>  |
|            | node_count      | 90.2 | <b>+9.0</b>  | <b>+8.8</b>  | <b>+9.2</b>  | <b>+8.0</b>  | <b>+9.2</b>  | <b>+9.8</b>  |
|            | node_degree     | 99.2 | +0.2         | -0.5         | <b>-2.2</b>  | <b>-6.5</b>  | -0.5         | <b>-8.0</b>  |

Table 2: Main sweep, plain arms (thinking arms in Table 3),  $n=40$ , pooled over densities  $p \in \{0.10, 0.20, 0.35, 0.50\}$ , 100 graphs per (task, primer, density) cell. The none column is absolute accuracy (%); all other columns are paired differences against none in percentage points. Truncated generations excluded pairwise,  $n \in [211, 400]$  paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family,  $q = 0.05$ .

| Arm              | Task            | none  | comp. | clust. | rwse        | degree       | filler | all          |
|------------------|-----------------|-------|-------|--------|-------------|--------------|--------|--------------|
| qwen3-1.7b-think | connected_nodes | 79.7  | +2.8  | +1.5   | -0.3        | <b>+7.4</b>  | +2.0   | <b>+6.1</b>  |
|                  | cycle_check     | 100.0 | +0.0  | +0.0   | +0.0        | -0.7         | +0.0   | +0.0         |
|                  | edge_count      | 86.3  | +7.4  | -8.7   | -4.0        | <b>-61.5</b> | +3.4   | -8.3         |
|                  | edge_existence  | 99.2  | -0.8  | -1.0   | -1.3        | -2.3         | -0.8   | -2.0         |
|                  | node_count      | 98.6  | +0.9  | +1.4   | +1.4        | +0.8         | +1.4   | +1.4         |
|                  | node_degree     | 76.8  | -1.0  | +1.0   | +2.5        | <b>+11.7</b> | -3.5   | <b>+11.6</b> |
| qwen3-4b-think   | connected_nodes | 98.7  | +0.5  | +0.0   | +0.0        | +0.0         | +0.8   | -0.5         |
|                  | cycle_check     | 100.0 | +0.0  | +0.0   | +0.0        | -0.8         | +0.0   | +0.0         |
|                  | edge_count      | 97.6  | +1.7  | +1.7   | +0.0        | <b>-23.5</b> | -2.0   | <b>-41.5</b> |
|                  | edge_existence  | 100.0 | +0.0  | +0.0   | +0.0        | +0.0         | +0.0   | +0.0         |
|                  | node_count      | 99.5  | +0.0  | +0.5   | +0.5        | +0.5         | +0.5   | +0.5         |
|                  | node_degree     | 99.7  | -0.3  | -0.3   | <b>-2.1</b> | +0.0         | +0.0   | -0.8         |

Table 3: Main sweep, thinking arms; conventions as in Table 2. Both arms are at or near ceiling on most tasks, leaving no headroom for a primer effect. edge\_count truncates on 62–81% of rows under thinking (Table 5) and is reported but not interpreted.  $n \in [13, 400]$  paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family,  $q = 0.05$ .

where our data are sparsest. Manipulating difficulty directly gives the same picture (Figure 3): for qwen3-1.7b-think on node\_degree, raising density from  $p=.10$  to  $.85$  lowers none accuracy from 0.95 to 0.44 while degree’s benefit rises from +3.1 to +34.2 points ( $r=-0.975$ ).

**Held-out arms and renaming.** On the eleven held-out arms the interaction replicates ( $-24.7$ , CI  $[-109, -3.8]$ , 43 (arm, task) blocks; route  $r=-0.674$  over 105 cells, no-route  $r=-0.186$  over 93). Replacing each cell’s own baseline with the other arms’ accuracy on the same cell removes the relation ( $r=+0.113$ ,  $p=0.135$ ), so the effect tracks a model’s own competence rather than item difficulty. Replacing integer node identifiers with Game-of-Thrones names leaves degree’s effect intact ( $r=0.933$  between the two sweeps over 39 cells), so the effect does not depend on integer labels; it cannot distinguish copying the stated value

from computing the answer. When the model errs despite a stated answer, the error is often a neighbour’s value: 12 of qwen3-4b’s 29 degree errors on node\_degree equal the stated degree of the adjacent node in the list (17.3% expected by chance,  $p=.002$ ).

#### 5.4 Non-answer content: clustering on node\_degree

clustering on node\_degree is the only cell in which a primer that does not state the answer was followed up; it was selected after the main sweep and re-run with 400 graphs per density and an independent-seed replication (Figure 4; Appendix Tables 10 and 11). Pooled over seven densities, the gain over none is small (+1.9 points,  $p=0.022$ ; replication at  $p \leq .50$ : +4.2,  $p=0.00055$ ). The larger pooled gain over filler (+4.3) mostly reflects filler’s own cost, which accounts for

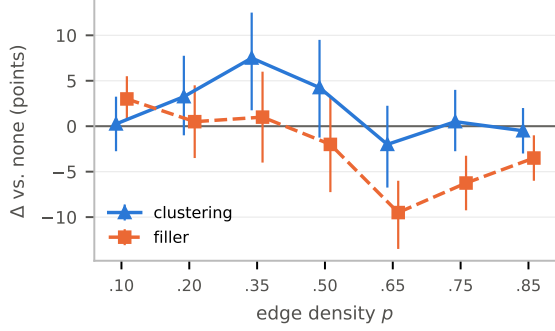


Figure 4: clustering and filler against none on node\_degree, qwen3-1.7b, by density (400 graphs per density, 95% bootstrap intervals). Pooled comparisons: Appendix Table 10.

the whole difference at  $p \geq .65$ . Against both controls, the effect is concentrated at intermediate densities: pooled over  $p = .35$  and  $.50$ , clustering exceeds none by +5.9 points (95% CI [2.1, 9.8],  $p = 0.0036$ ) and filler by +6.4 ([2.8, 10.0],  $p = 0.0012$ ), while filler and none do not differ (-0.5), so length does not account for it. This range was identified from the dedicated sweep; the independent-seed replication, which played no part in choosing it, gives +5.5 against none over the same two densities ([1.5, 9.5],  $p = 0.0083$ ). Per-density estimates are too imprecise to localise the effect more finely. qwen3-4b, at ceiling in the main sweep, shows the effect on the high-density extension (+11.3 against none, +9.3 against filler, both significant under the global correction); the thinking arms do not.

Neither mechanism we can test explains the effect (Appendix Figure 5). Under rewiring, mean clustering rises from 0.01 to 0.73 but the benefit does not follow (qwen3-1.7b: +3.5, +6.0, +2.5 across levels), and for qwen35-2b filler matches or exceeds clustering at every level. The density-prior prediction fails too: signed error under clustering should shift by up to 17 at high, but the paired shift is within 0.3 of zero at every level for all three arms (Appendix Table 12).

### 5.5 Thinking mode and power

Thinking brings most cells to ceiling, and where it does not, truncation censors the task. Rescoring the thinking arms at the plain arms' 2,048-token budget (exact under greedy decoding) shows that part of thinking's advantage is budget: it still wins on edge\_existence (97.8 vs. 70.5% for 1.7B) but loses on cycle\_check (38.0 vs. 90.7%),

and no edge\_count trace finishes (Appendix Table 13). Many main-sweep nulls are uninformative: in 10 of 24 null cells for qwen3-1.7b, 10/20 for qwen3-4b, 23/33 and 33/33 for the thinking arms, an effect in at least one direction is undetectable at any size because none accuracy is at floor or ceiling; elsewhere the median MDE for a gain is 5.6 (1.7B) and 10.0 points (4B).

## 6 Conclusion

Structural primers change LLM accuracy on GraphQA in two ways. For RQ1, a primer that states the requested quantity acts as a second route to the answer, helping a model at floor and harming one at ceiling; the contrast survives cross-fitting, a logit transformation, held-out arms, a negative-control difficulty estimate and node renaming, but is not graded at intermediate baselines. For RQ2, added text is an active treatment whose cost exceeds most content effects and grows with difficulty, so a primer compared only against none confounds content with length. For RQ3, the one surviving effect of non-answer content is small, concentrated at intermediate densities and unexplained. All claims are behavioural. Evaluations of prompt-level graph augmentations should report baseline accuracy, a length-matched control and a graph-blind solver.

## Limitations

The main sweep covers two model sizes from one family; the held-out arms extend the range but on a smaller corpus with fewer conditions and were not pre-registered. The route classification depends on a threshold and on the  $n=40$  shortcut refit, and the route relation is not detectable within the 0.10–0.90 baseline window, where data are sparse. All graphs are Erdős-Rényi, in which local clustering is nearly uniform, so the clustering primer carries little node-specific information; the rewiring varies triangle count within this family but does not test other families. The length control matches clustering to within 8% in tokens but is much shorter than rwse and all. Every prompt uses one encoding (incident), one template and one primer position, with greedy decoding. The clustering cell was selected after the main sweep, and several results rest on 400 or fewer instances per cell; an MDE is reported for every null result, but it does not replace a larger sample. Three primers would address the main

open questions: a *shuffled-value* primer (same values, permuted across nodes) as a stricter placebo than *filler*, a *corrupted* primer stating wrong degrees to separate copying from computation, and a *single-fact* primer to separate interference from route substitution.

## Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 200,000 generations.

## References

- Yoav Benjamini and Yoel Hochberg. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. The hitchhiker’s guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1383–1392.
- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2022. Graph neural networks with learnable structural and positional representations. In *International Conference on Learning Representations (ICLR)*.
- Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. 2023. Faith and fate: Limits of transformers on compositionality. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.
- Paul Erdős and Alfréd Rényi. 1959. On random graphs i. *Publicationes Mathematicae Debrecen*, 6:290–297.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2023. Talk like a graph: Encoding graphs for large language models. *arXiv preprint arXiv:2310.04560*.
- Bailey K. Fosdick, Daniel B. Larremore, Joel Nishimura, and Johan Ugander. 2018. Configuring random graph models with fixed degree sequences. *SIAM Review*, 60(2):315–355.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673.
- Andrew Gelman and Hal Stern. 2006. The difference between “significant” and “not significant” is not itself statistically significant. *The American Statistician*, 60(4):328–331.
- Google DeepMind. 2026. Gemma 4 model cards (gemma-4-E4B-it, gemma-4-12B-it). <https://huggingface.co/google>.
- Jiayan Guo, Lun Du, and Hengyu Liu. 2023. Gpt4graph: Can large language models understand graph structured data? an empirical evaluation and benchmarking. *arXiv preprint arXiv:2305.15066*.
- Michael J. Hautus. 1995. Corrections for extreme proportions and their biasing effects on estimated values of  $d'$ . *Behavior Research Methods, Instruments, & Computers*, 27(1):46–51.
- Bowen Jin, Gang Liu, Chi Han, Meng Jiang, Heng Ji, and Jiawei Han. 2024. Large language models on graphs: A comprehensive survey. *IEEE Transactions on Knowledge and Data Engineering*, 36(12):8622–8642.
- Greg Kamradt. 2023. Needle in a haystack – pressure testing llms. [https://github.com/gkamradt/LLMTest\\_NeedleInAHaystack](https://github.com/gkamradt/LLMTest_NeedleInAHaystack).
- Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. Same task, more tokens: the impact of input length on the reasoning performance of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 15339–15353.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Neil A. Macmillan and C. Douglas Creelman. 2005. *Detection Theory: A User’s Guide*, 2nd edition. Lawrence Erlbaum Associates.
- Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.
- Bryan Perozzi, Bahare Fatemi, Dustin Zelle, Anton Tsitsulin, Mehran Kazemi, Rami Al-Rfou, and Jonathan Halcrow. 2024. Let your graph do the talking: Encoding structured data for LLMs. *arXiv preprint arXiv:2402.05862*.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Qwen Team. 2026. Qwen3.5-2B model card. <https://huggingface.co/Qwen/Qwen3.5-2B>.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 31210–31227.

Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. Can language models solve graph problems in natural language? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35.

## A Supplementary material

### A.1 Example prompt

A clustering-condition node\_degree prompt at  $p=0.10$  (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.  
In this graph:  
Node 0 is connected to nodes 2, 6, 8, 12, 35.  
Node 1 is connected to nodes 5, 20.  
...  
Node 39 is connected to nodes 15, 16, 18, 20.  
Q: What is the degree of node 16?  
A:

### A.2 Robustness analyses

| Task            | none | comp.       | clust.      | rwse        | degree      | filler      | all         |
|-----------------|------|-------------|-------------|-------------|-------------|-------------|-------------|
| node_count      | .064 | .076        | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> | <b>1.00</b> |
| edge_count      | .018 | .018        | .148        | .018        | <b>1.00</b> | .018        | <b>1.00</b> |
| node_degree     | .082 | .082        | .082        | .616        | <b>1.00</b> | .082        | <b>1.00</b> |
| connected_nodes | .082 | .082        | .082        | .082        | .208        | .082        | .352        |
| edge_existence  | .498 | .498        | .720        | .642        | .794        | .498        | .794        |
| cycle_check     | .832 | <b>1.00</b> | .832        | .832        | .946        | .832        | .946        |

Table 4: Shortcut bars on 500 graphs from the published 5–19-node generator, used for the held-out arms. At  $n=40$  (Table 1) the edge\_existence/none bar rises to 0.735 through class imbalance, and degree’s gap over none on that task falls from +0.296 to +0.011.

| Task            | 1.7B  | 1.7B-th.     | 4B   | 4B-th.       |
|-----------------|-------|--------------|------|--------------|
| edge_count      | 22.4% | <b>81.0%</b> | 0.6% | <b>62.2%</b> |
| cycle_check     | 6.1%  | 6.5%         | 1.0% | 1.9%         |
| node_degree     | 0.0%  | 1.1%         | 0.0% | 1.6%         |
| connected_nodes | 0.0%  | 0.8%         | 0.0% | 3.5%         |
| edge_existence  | 0.0%  | 0.2%         | 0.1% | 0.8%         |
| node_count      | 0.0%  | 2.5%         | 0.0% | 0.1%         |

Table 5: Truncation rate (share of main-sweep generations reaching the output budget: 2,048 tokens plain, 8,192 thinking), by task and arm, pooled over condition and density. **Bold**: above 50%.

| Arm        | Cond.      | hit rate | false alarm |
|------------|------------|----------|-------------|
| qwen3-1.7b | none       | 100.0%   | 40.3%       |
|            | filler     | 100.0%   | 61.8%       |
|            | clustering | 100.0%   | 35.2%       |
| qwen3-4b   | none       | 99.1%    | 12.6%       |
|            | filler     | 100.0%   | 22.5%       |
|            | clustering | 100.0%   | 7.5%        |

Table 6: edge\_existence,  $p=0.50$ : hit rate (correct on gold “yes”) and false-alarm rate (“yes” on gold “no”). Accuracy changes almost entirely through the false-alarm rate.

| Cond.      | qwen3-1.7b  |                |       | qwen3-4b    |                |       |
|------------|-------------|----------------|-------|-------------|----------------|-------|
|            | $\Delta d'$ | $\Delta c$     |       | $\Delta d'$ | $\Delta c$     |       |
| components | −0.36       | [−0.56, −0.21] | −0.18 | +0.07       | [−0.24, +0.33] | −0.10 |
| clustering | +0.21       | [+0.03, +0.41] | +0.11 | +0.43       | [+0.12, +0.75] | +0.08 |
| rwse       | +0.07       | [−0.19, +0.43] | +0.28 | −0.38       | [−0.68, −0.15] | −0.32 |
| degree     | +0.15       | [−0.05, +0.41] | +0.19 | −0.04       | [−0.33, +0.21] | −0.15 |
| filler     | −0.93       | [−1.16, −0.71] | −0.46 | −0.25       | [−0.54, +0.00] | −0.25 |
| all        | +0.55       | [+0.35, +0.77] | +0.27 | −0.36       | [−0.71, −0.11] | −0.31 |

Table 7: edge\_existence, main sweep: change in sensitivity  $d'$  and criterion  $c$  against none, averaged over the four densities (log-linear correction; 95% bootstrap CI for  $\Delta d'$ ). With the hit rate at ceiling,  $d'$  and  $c$  are not separately identifiable.

| Accuracy under none          | substitute route |          | no route |          |
|------------------------------|------------------|----------|----------|----------|
|                              | cells            | $\Delta$ | cells    | $\Delta$ |
| 0.00–0.25                    | 24               | +6.3     | 24       | −0.7     |
| 0.25–0.50                    | 6                | +8.5     | 6        | +2.0     |
| 0.50–0.75                    | 6                | +15.5    | 24       | +0.4     |
| 0.75–0.95                    | 11               | +4.2     | 33       | −3.9     |
| 0.95–1.00                    | 34               | −9.0     | 186      | −0.4     |
| $r(\Delta, \text{baseline})$ | −0.417           |          | −0.050   |          |

Table 8: Mean primer effect (percentage points), binned by none accuracy on the same (arm, task, density, condition) cell of the main sweep, using the  $n=40$  route split (Table 1). Mean baseline is 0.61 for route cells and 0.85 for no-route cells, so the groups are not baseline-matched; the cross-fitted interaction test in the text addresses this.

| Estimate                      | $r_{\text{route}}$ | $k$ | $r_{\text{none}}$ | $k$ | interaction [95% CI] |
|-------------------------------|--------------------|-----|-------------------|-----|----------------------|
| naive, pp                     | -0.35              | 117 | -0.01             | 381 | -12.5 [-19.5, -6.3]  |
| cross-fit (one direction), pp | -0.34              | 116 | +0.03             | 381 | -12.1 [-17.9, -7.0]  |
| cross-fit (both), pp          | -0.31              | 232 | +0.02             | 762 | -11.0 [-17.1, -5.7]  |
| cross-fit (both), logit       | -0.40              | 232 | +0.04             | 762 | -1.35 [-1.93, -0.77] |
| baseline .10-.90, pp          | -0.01              | 78  | -0.07             | 187 | +3.0 [-29.1, +41.1]  |
| baseline .10-.90, logit       | +0.02              | 78  | -0.05             | 187 | +0.22 [-1.36, +1.94] |
| held-out, pp                  | -0.67              | 105 | -0.19             | 93  | -24.7 [-109, -3.8]   |

Table 9: Route  $\times$  baseline relation, main sweep plus high-density extension (held-out: published split and probe). “Both” counts each cell once per fold direction;  $k$  for the window rows counts points. Interaction CIs resample (arm, task, density) blocks, or (arm, task) for held-out arms.

| Comparison                 | $\Delta$ | $n$  | density | $p$                  |
|----------------------------|----------|------|---------|----------------------|
| clust. vs. none (main)     | +3.0     | 400  | .10-.50 | 0.26                 |
| clust. vs. none            | +1.9     | 2800 | .10-.85 | 0.022                |
| filler vs. none            | -2.4     | 2800 | .10-.85 | 0.0013               |
| clust. vs. filler          | +4.3     | 2800 | .10-.85 | $2.4 \times 10^{-8}$ |
| clust. vs. none            | +7.5     | 400  | .35     | 0.0089               |
| clust. vs. none (new seed) | +4.2     | 1600 | .10-.50 | 0.00055              |

Table 10: qwen3-1.7b, node\_degree. All rows except the first come from the dedicated 400-graph-per-density sweep; the last row uses newly seeded graphs. Per-density values are in Appendix Table 11.

| $p$ | clust.-none | filler-none | clust.-filler |
|-----|-------------|-------------|---------------|
| .10 | +0.3        | +3.0*       | -2.8*         |
| .20 | +3.3        | +0.5        | +2.8          |
| .35 | +7.5*       | +1.0        | +6.5*         |
| .50 | +4.3        | -2.0        | +6.3*         |
| .65 | -2.0        | -9.5*       | +7.5*         |
| .75 | +0.5        | -6.3*       | +6.8*         |
| .85 | -0.5        | -3.5*       | +3.0*         |

Table 11: qwen3-1.7b, node\_degree, dedicated sweep, 400 graphs per density. \*: McNemar  $p < 0.05$  (uncorrected).

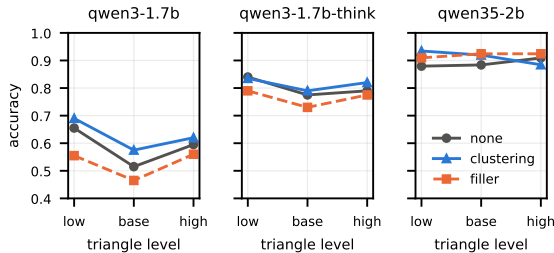


Figure 5: Rewiring: node\_degree accuracy by triangle level (200 graphs per level; degree sequence and prompt length fixed; mean clustering 0.01, 0.31, 0.73).

| Arm              | low   | base  | high  |
|------------------|-------|-------|-------|
| predicted shift  | -11.5 | 0.0   | +16.7 |
| qwen3-1.7b       | +0.03 | +0.28 | -0.02 |
| qwen3-1.7b-think | +0.01 | +0.04 | +0.01 |
| qwen35-2b        | -0.29 | -0.26 | -0.16 |

Table 12: Density-prior test on the rewiring corpus: mean paired shift in signed node\_degree error, clustering minus none (errors clipped at  $\pm 20$ , 200 graphs per level). “Predicted” is the shift if answers moved fully to  $\bar{C}(n-1)$ .

| Task            | 1.7B, acc. under none (%) |       |            | 4B, acc. under none (%) |       |            |
|-----------------|---------------------------|-------|------------|-------------------------|-------|------------|
|                 | plain                     | think | think@2048 | plain                   | think | think@2048 |
| connected_nodes | 72.5                      | 78.7  | 67.2       | 86.0                    | 95.5  | 89.0       |
| cycle_check     | 90.7                      | 97.0  | 38.0       | 99.0                    | 99.3  | 62.0       |
| edge_count      | 1.3                       | 15.8  | 0.0        | 2.0                     | 20.0  | 0.0        |
| edge_existence  | 70.5                      | 99.0  | 97.8       | 90.5                    | 99.8  | 99.8       |
| node_count      | 31.5                      | 91.0  | 86.0       | 90.2                    | 99.5  | 94.8       |
| node_degree     | 60.5                      | 76.2  | 67.7       | 99.3                    | 97.0  | 91.2       |

Table 13: Budget-matched comparison. Plain arms use 2,048 tokens; thinking arms 8,192 (“think”) or rescored at 2,048 (“think@2048”). Truncated generations are scored incorrect here, so values differ from the dropped-pair convention of Tables 2 and 3.

| Arm  | Task        | none  | comp. | clust. | rwse | degree | filler | all  |
|------|-------------|-------|-------|--------|------|--------|--------|------|
| 1.7B | node_count  | 0.69  | 0.47  | 0.01   | 0.67 | 0.62   | 0.29   | 0.22 |
|      | edge_count  | 133.4 | 98.1  | 105.7  | 79.0 | 47.7   | 181.2  | 94.2 |
|      | node_degree | 0.94  | 0.91  | 0.78   | 0.91 | 0.67   | 1.27   | 0.91 |
| 4B   | node_count  | 0.10  | 0.01  | 0.01   | 0.01 | 0.02   | 0.01   | 0.00 |
|      | edge_count  | 39.2  | 40.1  | 43.4   | 56.0 | 40.4   | 29.3   | 28.1 |
|      | node_degree | 0.02  | 0.01  | 0.02   | 0.05 | 0.14   | 0.02   | 0.49 |

Table 14: Mean absolute error on the integer tasks, plain arms, main sweep (non-truncated rows).

### A.3 Per-cell results

Paired effect sizes for every (arm, task, primer) cell:  $\Delta$  in percentage points against none, a 95% paired-bootstrap interval, the permutation  $p$ -value, and  $\Delta$  under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 3.4).  $n$  is the number of pairs in which neither arm truncated.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 400 | +0.8     | [-2.8, +4.5]   | 0.7861   | +0.8   |
|                 | clustering | 400 | +1.2     | [-3.0, +5.8]   | 0.6458   | +1.2   |
|                 | rwse       | 400 | +1.0     | [-3.5, +5.2]   | 0.7304   | +1.0   |
|                 | degree     | 400 | +3.5     | [-0.5, +7.5]   | 0.1239   | +3.5   |
|                 | filler     | 400 | -6.2     | [-10.2, -2.2]  | 0.0040   | -6.2   |
|                 | all        | 399 | +2.0     | [-2.8, +6.8]   | 0.4644   | +2.0   |
| cycle_check     | components | 343 | -0.6     | [-1.5, +0.0]   | 0.5033   | +2.0   |
|                 | clustering | 362 | +0.3     | [+0.0, +0.8]   | 1.0000   | +8.5   |
|                 | rwse       | 358 | -7.5     | [-10.3, -4.7]  | 0.0001   | -0.2   |
|                 | degree     | 301 | -12.6    | [-16.6, -9.0]  | 0.0001   | -17.5  |
|                 | filler     | 341 | +0.3     | [+0.0, +0.9]   | 1.0000   | +3.0   |
|                 | all        | 362 | -1.1     | [-2.5, +0.0]   | 0.2210   | +7.0   |
| edge_count      | components | 211 | -1.4     | [-3.8, +0.5]   | 0.3783   | -1.0   |
|                 | clustering | 238 | -0.4     | [-2.5, +1.7]   | 1.0000   | -0.5   |
|                 | rwse       | 239 | -0.8     | [-2.9, +1.3]   | 0.6837   | -0.8   |
|                 | degree     | 253 | +4.0     | [+0.8, +7.1]   | 0.0179   | +3.0   |
|                 | filler     | 224 | -0.4     | [-2.2, +1.3]   | 1.0000   | -0.5   |
|                 | all        | 211 | +2.4     | [-0.5, +5.7]   | 0.2269   | +0.8   |
| edge_existence  | components | 400 | -8.2     | [-11.8, -4.8]  | 0.0001   | -8.2   |
|                 | clustering | 400 | +3.8     | [-0.5, +8.0]   | 0.1199   | +3.8   |
|                 | rwse       | 399 | +5.3     | [+0.8, +9.8]   | 0.0321   | +5.0   |
|                 | degree     | 400 | +4.0     | [-0.5, +8.2]   | 0.0957   | +4.0   |
|                 | filler     | 400 | -15.8    | [-20.0, -11.8] | 0.0001   | -15.8  |
|                 | all        | 400 | +11.5    | [+6.8, +16.2]  | 0.0001   | +11.5  |
| node_count      | components | 400 | +21.2    | [+15.8, +26.8] | 0.0001   | +21.2  |
|                 | clustering | 400 | +67.2    | [+62.7, +71.8] | 0.0001   | +67.2  |
|                 | rwse       | 400 | +2.0     | [-4.2, +8.5]   | 0.5871   | +2.0   |
|                 | degree     | 400 | +6.8     | [+3.0, +10.5]  | 0.0008   | +6.8   |
|                 | filler     | 400 | +39.8    | [+35.0, +44.8] | 0.0001   | +39.8  |
|                 | all        | 400 | +47.0    | [+42.0, +52.0] | 0.0001   | +47.0  |
| node_degree     | components | 400 | +1.8     | [-2.0, +5.8]   | 0.4599   | +1.8   |
|                 | clustering | 400 | +3.0     | [-1.8, +7.8]   | 0.2577   | +3.0   |
|                 | rwse       | 399 | +4.0     | [-0.5, +8.5]   | 0.1104   | +4.0   |
|                 | degree     | 400 | +7.5     | [+3.0, +12.0]  | 0.0018   | +7.5   |
|                 | filler     | 400 | -0.5     | [-4.5, +3.5]   | 0.9028   | -0.5   |
|                 | all        | 400 | +10.2    | [+5.2, +15.2]  | 0.0003   | +10.2  |

Table 15: qwen3-1.7b: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 392 | +2.8     | [-0.5, +6.1]   | 0.1268   | +2.5   |
|                 | clustering | 394 | +1.5     | [-2.5, +5.6]   | 0.5384   | +2.0   |
|                 | rwse       | 393 | -0.3     | [-4.1, +3.6]   | 1.0000   | +0.2   |
|                 | degree     | 390 | +7.4     | [+3.6, +11.3]  | 0.0001   | +6.2   |
|                 | filler     | 394 | +2.0     | [-1.5, +5.6]   | 0.3275   | +2.5   |
|                 | all        | 392 | +6.1     | [+2.6, +9.9]   | 0.0027   | +6.2   |
| cycle_check     | components | 346 | +0.0     | [+0.0, +0.0]   | 1.0000   | -7.5   |
|                 | clustering | 376 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.5   |
|                 | rwse       | 386 | +0.0     | [+0.0, +0.0]   | 1.0000   | +2.2   |
|                 | degree     | 295 | -0.7     | [-1.7, +0.0]   | 0.5036   | -21.8  |
|                 | filler     | 388 | +0.0     | [+0.0, +0.0]   | 1.0000   | +3.0   |
|                 | all        | 373 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.8   |
| edge_count      | components | 27  | +7.4     | [+0.0, +18.5]  | 0.5037   | -0.8   |
|                 | clustering | 23  | -8.7     | [-26.1, +8.7]  | 0.6251   | -1.2   |
|                 | rwse       | 25  | -4.0     | [-12.0, +0.0]  | 1.0000   | -3.2   |
|                 | degree     | 13  | -61.5    | [-84.6, -38.5] | 0.0102   | -3.2   |
|                 | filler     | 29  | +3.4     | [-6.9, +13.8]  | 1.0000   | -1.5   |
|                 | all        | 24  | -8.3     | [-29.2, +12.5] | 0.6794   | +5.0   |
| edge_existence  | components | 398 | -0.8     | [-2.3, +0.8]   | 0.5090   | -0.8   |
|                 | clustering | 398 | -1.0     | [-2.5, +0.3]   | 0.2911   | -1.0   |
|                 | rwse       | 398 | -1.3     | [-3.0, +0.3]   | 0.2274   | -1.2   |
|                 | degree     | 397 | -2.3     | [-4.3, -0.5]   | 0.0249   | -2.5   |
|                 | filler     | 399 | -0.8     | [-2.3, +0.8]   | 0.5051   | -0.5   |
|                 | all        | 399 | -2.0     | [-3.8, -0.5]   | 0.0374   | -1.8   |
| node_count      | components | 342 | +0.9     | [-0.3, +2.3]   | 0.3758   | +1.5   |
|                 | clustering | 369 | +1.4     | [+0.3, +2.7]   | 0.0597   | +9.0   |
|                 | rwse       | 369 | +1.4     | [+0.3, +2.7]   | 0.0597   | +9.0   |
|                 | degree     | 360 | +0.8     | [-0.6, +2.2]   | 0.4512   | +5.5   |
|                 | filler     | 369 | +1.4     | [+0.3, +2.7]   | 0.0597   | +9.0   |
|                 | all        | 369 | +1.4     | [+0.3, +2.7]   | 0.0597   | +9.0   |
| node_degree     | components | 395 | -1.0     | [-4.3, +2.3]   | 0.6543   | -1.0   |
|                 | clustering | 396 | +1.0     | [-2.8, +4.8]   | 0.7010   | +1.5   |
|                 | rwse       | 394 | +2.5     | [-1.5, +6.9]   | 0.2898   | +2.5   |
|                 | degree     | 376 | +11.7    | [+7.4, +16.0]  | 0.0001   | +8.8   |
|                 | filler     | 396 | -3.5     | [-7.3, +0.3]   | 0.0954   | -3.0   |
|                 | all        | 396 | +11.6    | [+7.3, +15.9]  | 0.0001   | +12.2  |

Table 16: qwen3-1.7b-think: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 400 | +9.2     | [+6.0, +12.8]  | 0.0001   | +9.2   |
|                 | clustering | 400 | +1.0     | [-3.0, +4.8]   | 0.6893   | +1.0   |
|                 | rwse       | 400 | +0.0     | [-4.0, +4.0]   | 1.0000   | +0.0   |
|                 | degree     | 400 | -3.0     | [-7.2, +1.2]   | 0.1882   | -3.0   |
|                 | filler     | 400 | -12.5    | [-17.0, -8.0]  | 0.0001   | -12.5  |
|                 | all        | 400 | -2.8     | [-7.0, +1.5]   | 0.2511   | -2.8   |
| cycle_check     | components | 391 | -1.3     | [-2.6, +0.0]   | 0.1318   | -2.0   |
|                 | clustering | 395 | +0.3     | [+0.0, +0.8]   | 1.0000   | +0.5   |
|                 | rwse       | 392 | +0.3     | [+0.0, +0.8]   | 1.0000   | -0.2   |
|                 | degree     | 391 | -1.5     | [-2.8, -0.5]   | 0.0312   | -2.2   |
|                 | filler     | 392 | -0.3     | [-0.8, +0.0]   | 1.0000   | -0.5   |
|                 | all        | 395 | -0.5     | [-1.5, +0.5]   | 0.6269   | -0.2   |
| edge_count      | components | 397 | -1.3     | [-2.8, +0.3]   | 0.1756   | -1.2   |
|                 | clustering | 399 | -1.3     | [-3.0, +0.3]   | 0.2298   | -1.2   |
|                 | rwse       | 398 | -1.5     | [-3.0, -0.3]   | 0.0668   | -1.5   |
|                 | degree     | 393 | +27.7    | [+23.2, +32.6] | 0.0001   | +27.3  |
|                 | filler     | 400 | +1.2     | [-0.5, +3.2]   | 0.3012   | +1.2   |
|                 | all        | 397 | +3.8     | [+1.3, +6.3]   | 0.0049   | +3.8   |
| edge_existence  | components | 400 | -2.8     | [-6.0, +0.5]   | 0.1254   | -2.8   |
|                 | clustering | 400 | +4.0     | [+1.0, +7.0]   | 0.0168   | +4.0   |
|                 | rwse       | 400 | -10.0    | [-14.0, -6.2]  | 0.0001   | -10.0  |
|                 | degree     | 398 | -4.3     | [-7.8, -1.0]   | 0.0200   | -4.8   |
|                 | filler     | 400 | -7.0     | [-10.8, -3.5]  | 0.0002   | -7.0   |
|                 | all        | 399 | -9.8     | [-13.8, -6.0]  | 0.0001   | -10.0  |
| node_count      | components | 400 | +9.0     | [+6.2, +12.0]  | 0.0001   | +9.0   |
|                 | clustering | 400 | +8.8     | [+5.8, +11.8]  | 0.0001   | +8.8   |
|                 | rwse       | 400 | +9.2     | [+6.2, +12.2]  | 0.0001   | +9.2   |
|                 | degree     | 400 | +8.0     | [+5.0, +11.2]  | 0.0001   | +8.0   |
|                 | filler     | 400 | +9.2     | [+6.5, +12.2]  | 0.0001   | +9.2   |
|                 | all        | 400 | +9.8     | [+7.0, +12.8]  | 0.0001   | +9.8   |
| node_degree     | components | 400 | +0.2     | [+0.0, +0.8]   | 1.0000   | +0.2   |
|                 | clustering | 400 | -0.5     | [-2.0, +0.8]   | 0.7246   | -0.5   |
|                 | rwse       | 400 | -2.2     | [-4.0, -0.5]   | 0.0240   | -2.2   |
|                 | degree     | 400 | -6.5     | [-9.2, -3.8]   | 0.0001   | -6.5   |
|                 | filler     | 400 | -0.5     | [-2.0, +0.8]   | 0.7232   | -0.5   |
|                 | all        | 400 | -8.0     | [-10.8, -5.2]  | 0.0001   | -8.0   |

Table 17: qwen3-4b: all cells.

| Task            | Primer     | $n$ | $\Delta$ | 95% CI         | perm $p$ | zeroed |
|-----------------|------------|-----|----------|----------------|----------|--------|
| connected_nodes | components | 377 | +0.5     | [-0.5, +1.6]   | 0.6249   | +0.5   |
|                 | clustering | 373 | +0.0     | [-1.3, +1.3]   | 1.0000   | -0.8   |
|                 | rwse       | 379 | +0.0     | [-1.3, +1.6]   | 1.0000   | +0.5   |
|                 | degree     | 376 | +0.0     | [-1.6, +1.6]   | 1.0000   | +0.5   |
|                 | filler     | 379 | +0.8     | [-0.3, +2.1]   | 0.3787   | +1.2   |
|                 | all        | 375 | -0.5     | [-2.1, +0.8]   | 0.7306   | -1.2   |
| cycle_check     | components | 381 | +0.0     | [+0.0, +0.0]   | 1.0000   | -3.2   |
|                 | clustering | 394 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.0   |
|                 | rwse       | 395 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.2   |
|                 | degree     | 375 | -0.8     | [-1.9, +0.0]   | 0.2514   | -5.5   |
|                 | filler     | 396 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.5   |
|                 | all        | 392 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.5   |
| edge_count      | components | 60  | +1.7     | [+0.0, +5.0]   | 1.0000   | +24.8  |
|                 | clustering | 59  | +1.7     | [-3.4, +6.8]   | 1.0000   | +24.8  |
|                 | rwse       | 49  | +0.0     | [-6.1, +6.1]   | 1.0000   | +10.2  |
|                 | degree     | 34  | -23.5    | [-38.2, -8.8]  | 0.0073   | +8.5   |
|                 | filler     | 50  | -2.0     | [-6.0, +0.0]   | 1.0000   | +12.2  |
|                 | all        | 41  | -41.5    | [-56.1, -26.8] | 0.0002   | +1.8   |
| edge_existence  | components | 396 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.5   |
|                 | clustering | 396 | +0.0     | [+0.0, +0.0]   | 1.0000   | -0.8   |
|                 | rwse       | 393 | +0.0     | [+0.0, +0.0]   | 1.0000   | -1.2   |
|                 | degree     | 398 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.0   |
|                 | filler     | 399 | +0.0     | [+0.0, +0.0]   | 1.0000   | +0.0   |
|                 | all        | 394 | +0.0     | [+0.0, +0.0]   | 1.0000   | -1.0   |
| node_count      | components | 400 | +0.0     | [-1.0, +1.0]   | 1.0000   | +0.0   |
|                 | clustering | 399 | +0.5     | [+0.0, +1.3]   | 0.4936   | +0.2   |
|                 | rwse       | 399 | +0.5     | [+0.0, +1.3]   | 0.5005   | +0.2   |
|                 | degree     | 399 | +0.5     | [+0.0, +1.3]   | 0.5005   | +0.2   |
|                 | filler     | 400 | +0.5     | [+0.0, +1.2]   | 0.4870   | +0.5   |
|                 | all        | 400 | +0.5     | [+0.0, +1.2]   | 0.4870   | +0.5   |
| node_degree     | components | 380 | -0.3     | [-0.8, +0.0]   | 1.0000   | +0.2   |
|                 | clustering | 386 | -0.3     | [-1.0, +0.5]   | 1.0000   | +1.8   |
|                 | rwse       | 381 | -2.1     | [-3.7, -0.8]   | 0.0072   | -1.2   |
|                 | degree     | 388 | +0.0     | [-0.8, +0.8]   | 1.0000   | +2.5   |
|                 | filler     | 386 | +0.0     | [+0.0, +0.0]   | 1.0000   | +2.0   |
|                 | all        | 383 | -0.8     | [-1.8, +0.0]   | 0.2466   | +0.5   |

Table 18: qwen3-4b-think: all cells.